

How Detectors Find Candidate Object Locations

Computer Vision

R-CNN

RPN

Object Detection

Anchor Boxes

What Is a Region Proposal?

A region proposal (or region of interest, RoI) is a candidate bounding box that might contain an object. Instead of scanning every possible window in an image (brute force), proposal methods quickly generate a smaller set of high-likelihood candidate regions that a classifier then evaluates. This two-stage design dramatically improves both speed and accuracy compared to exhaustive sliding window search.

Relationship to RAG / LangChain / NLP

- **Pure Computer Vision concept** — Region proposals are specific to object detection pipelines, not related to NLP, RAG, or LangChain.
- **Agentic AI vision tools** — When building LLM agents with tool use for vision, an object detector (which uses proposals internally) can be a tool called by LangChain/LangGraph.
- **Document parsing** — Layout detection models (used to parse PDFs/invoices for RAG) use region proposals to locate text blocks and tables.

Evolution: From Selective Search to RPN

Method	Year	How It Generates Proposals	Speed
Sliding Window	Pre-2014	Dense multi-scale window scan	Very slow (millions of windows)
Selective Search	2013	Group similar pixels (colour, texture) into hierarchical regions	~2s/image (CPU)
EdgeBoxes	2014	Score boxes by edge density inside them	~0.2s/image
RPN (Faster RCNN)	2015	Fully conv net shares backbone with detector; predicts anchors	~10ms on GPU
Anchor-Free	2019+	Predict center points (CenterNet, FCOS) instead of anchors	Fastest

Region Proposal Network (RPN) — Deep Dive

Introduced in Faster R-CNN (Ren et al., 2015), the RPN is a small fully-convolutional network that slides over the shared feature map. At each spatial location it evaluates k anchor boxes (combinations of scales $\times$ aspect-ratios). For each anchor it outputs two things: an objectness score (object vs background) and four box-offset deltas.

- **Anchors** — Pre-defined reference boxes. Typical: 3 scales × 3 ratios = 9 anchors per location.
- **Objectness score** — Binary classifier: does this anchor overlap an object (IoU > 0.7 → positive)?
- **Box regression** — Predict (Δx , Δy , Δw , Δh) offsets to refine the anchor into a tight proposal.
- **NMS on proposals** — Top-N proposals (e.g. 300) kept after NMS before the detection head.

RoI Pooling & RoI Align

After proposals are generated, each RoI must be mapped to a fixed-size feature map for the classifier head. RoI Pooling (Fast R-CNN) does this with max-pool but introduces quantisation misalignment. RoI Align (Mask R-CNN) fixes this with bilinear interpolation — critical for tasks like instance segmentation.

Modern Trend

Transformer-based detectors (DETR and successors) eliminate region proposals entirely, instead treating detection as a direct set prediction problem using cross-attention between query vectors and image features. This simplifies the pipeline but requires more data/training.

The Full Faster R-CNN Pipeline

Backbone (ResNet) → FPN → RPN → RoI Align → Detection Head → NMS → Output

- **Backbone** — CNN (e.g. ResNet-50) extracts feature maps from the image.
- **FPN (Feature Pyramid Network)** — Multi-scale feature maps for detecting small and large objects.
- **RPN** — Generates ~2000 candidate proposals per image.
- **RoI Align** — Warps each proposal region to 7×7 feature grid.
- **Detection Head** — FC layers output final class + refined box coordinates.